

STRATEGIC PLANING FOR DATA DRIVEN LOGISTICS OPTIMIZATION

PROJECT 3 – Exploratory Data Analysis and
Visualisation of Delivery Logistics Data

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DATASET USED – DELIVERY LOGISTICS DATA

RECORDS ANALYSED – 25,000

PRIMARY BUSINESS PROBLEM – Reduce
anamolies in data, to reduce any kind of
misinterpretation in future business decisions.

TOOLS USED – Python, Pandas, Pycharm,
Matplotlib

EXECUTIVE SUMMARY

This project is about the investigation and analysis of logistics data via Python programming. Its aim is to discover correlations in delivery processes and convert them into usable business findings.

A dataset entitled Delivery_Logistics was used for the analysis which consists of 25,000 instances. Exploratory Data Analysis was used to study the group means and dispersions, check relations and correlations among important logistic variables like delivery distance, parcel weight and cost, customer rating, delivery type, weather conditions, and delivery result.

The numeric analysis shows that the mean delivery distance equals 150.39 km, the mean weight of parcel is 25.15 kg, the mean cost of delivery is 864.94, and the mean customer rating equals 3.67.

The correlation analysis shows that there is a strong positive correlation between the delivery distance and delivery cost ($r = 0.991$). The connection between the customer rating and delivery cost is moderate and negative ($r = -0.161$) as well as between the customer rating and delivery distance ($r = -0.133$).

By making categorical comparisons, we can see several significant operational variations. The most significant differences are in the delays depending on the type of delivery service. For example, there is a delay rate of 73.78% for Express deliveries, compared to 0.43% for Two-Day deliveries and 0% for Standard deliveries. Weather conditions also show significant variations, with Stormy weather resulting in a delay rate of 41.45% and Cold weather resulting in 16.02%.

We will use Python for our visualizations to present the results of our analysis in terms of distributions, comparisons, and relationships among different parameters. The outcomes of this analysis are expected to help the logistics managers to identify cost factors, operational obstacles, and the areas which should be examined in more detail.

2. Introduction

Logistics companies produce massive numbers of operation-related data. However, it is crucial to be able to determine useful patterns from the data to have it useful in practice.

Exploratory Data Analysis allows us to analyse a dataset in a systematic way and understand it before running further analysis and predictions. It involves the numerical analysis backed by the graphical presentation of the information in order to analyse the distributions and correlations.

This project uses EDA to analyse the Delivery_Logistics dataset used in the previous projects. In this process:

- delivery distance,
- weight of packages,
- cost of delivery,
- customer rating,
- mode of delivery,
- weather conditions,
- state of delivery,
- delay in delivery, and
- relationship between different variables are analysed.

The Python libraries, such as Pandas, Matplotlib, Seaborn, and Scikit-learn are used for this task.

The process of analysis is as follows: Dataset → Descriptive Statistics → Distribution Analysis → Categorical Analysis → Correlation Analysis → Visualization → Business Insights → Suggestions.

3. OBJECTIVES

The main objectives of Project 3 include:

- Performing a thorough exploratory data analysis of the logistics dataset.
- Calculating the various measures of central tendency such as mean and median.
- Analysing the distribution of the main numerical variables.
- Analyzing the association of the logistics variables through correlation analysis.
- Generating suitable visuals for performance metrics of logistics.
- Evaluating the delivery performance in terms of modes of delivery, delivery partners, geographic regions, and climatic conditions.
- Identifying possible bottlenecks and cost drivers.
- Discussing the rationale behind using each visualization technique.
- Deriving actionable logistics insights through the analysis.
- Debating the recommendations based on the findings from the dataset.

4. EDS Methodology and Dataset Preparation

4.1 Dataset used

In Project 3 the same Delivery_Logistics dataset from Projects 1 and 2 is used. The employment of the same dataset across the projects facilitates the consistency of the study and the consecutive development of the analysis.

The original dataset includes 25,000 delivery records and comprises 15 variables. For Project 3 the version of the data prepared for analysis in Project 2 is applied.

The focus of the analysis is on the following variables:

- delivery_partner
- package_type
- vehicle_type
- delivery_mode
- region
- weather_condition
- distance_km
- package_weight_kg
- delayed
- delivery_status
- delivery_rating
- delivery_cost

The two unchanged variables from Project 2, delivery_time_hours and expected_time_hours, are removed from the analytic dataset because they do not have any variation.

4.2 EDA Methodology

The four step exploratory data analysis process was as follows:

Step 1: Descriptive statistics.

Certain statistics were derived for numerical variables such as:

- mean,
- median,
- minimum,
- maximum,
- standard deviation,
- quartiles.

Step 2: Distribution statistics.

Frequency distribution of data was viewed through the use of histogram and box plot charts for the determination of distribution of:

- delivery distance,
- package weight,
- delivery price,
- customer rating.

Step 3: Categorical data analysis.

Delivery performance across:

- delivery methods,
- delivery partners,
- region,
- weather conditions,
- vehicle types were performed and emphasis was given to delivery delays.

Step 4: Relationship and correlation analysis.

Correlation methods were employed to study the relationships between numerical logistics variables.

Major relations studied were:

- distance vs. delivery cost,
- package weight vs. delivery cost,
- distance vs. customer rating,
- delivery cost vs. customer rating,
- package weight vs. customer rating.

4.3 Python Setup

The following Python libraries were used for the analysis and visualization.

Python Code Illustration

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
# Load the original dataset
```

```
df = pd.read_csv("Delivery_Logistics(1).csv")
```

```
# Create analytical copy
```

```
df_eda = df.copy()
```

```
# Treat delivery ID as an identifier
```

```
df_eda["delivery_id"] = df_eda["delivery_id"].astype(str)
```

```
# Remove non-informative variables identified in Project 2
```

```
df_eda = df_eda.drop(  
    columns=["delivery_time_hours", "expected_time_hours"]  
)
```

```
print("EDA dataset shape:", df_eda.shape)
```

Result

The resulting EDA dataset contains:

25,000 records × 13 variables

No delivery records are removed during this preparation stage.

4.4 Numerical Variables Selected for EDA

The following numerical variables are the primary focus of the statistical analysis:

```
numeric_columns = [  
    "distance_km",  
    "package_weight_kg",  
    "delivery_rating",  
    "delivery_cost"  
]
```

```
print(df_eda[numeric_columns].describe())
```

These variables were selected because they represent important logistics performance and operational measures:

Variable	Logistics Meaning
distance_km	Transportation distance
package_weight_kg	Shipment weight
delivery_rating	Customer satisfaction
delivery_cost	Delivery expenditure

Table1

4.5 Why EDS is important for Logistics?

Exploratory Data Analysis converts unprocessed shipment data into valuable provided data to support managerial decision-making.

For instance, if the delivery fee depends on distances, the management can look at transportation efficiency and routing. If the delays are mostly in a specific deliver mode and weather condition, the operational capacity can be directed to that mode and weather.

In a similar manner, comparing the reviews from customers in various operational states can allow one to discover what factors correlate with customer enjoyment.

Thus the visualizations presented in the project will serve not just to make a report more appealing, each visualization has its own objective.

4.6 Visualization Selection Framework

Logistics Question	Visualization	Reason
What does delivery cost distribution look like?	Histogram	Shows distribution and concentration
Are there unusual numerical patterns?	Box plot	Shows spread and extreme observations

How does cost relate to distance?	Scatter plot	Shows relationship between two numerical variables
Which delivery modes have more delays?	Bar chart	Enables category comparison
Does weather affect delays?	Bar chart	Clearly compares delay rates
How are numerical variables related?	Correlation heatmap	Displays multiple relationships simultaneously
How do delivery partners compare?	Bar chart	Allows operational benchmarking

Table 2

This ensures that every visualization has an analytical purpose and directly addresses the requirements of the assignment.

Conclusion

The EDA method gives a systematic structure for delivery_Logistics data analysis. The analysis will include descriptive statistics, distributions, and comparisons along with correlations together with visualizations in order to find interesting patterns in delivery process operations.

5. Central Tendency and Descriptive Statistics

Descriptive statistics gives one an idea of the numerical features of the statistics of the logistics dataset. In this study, the mean, median, standard deviation, minimum, maximum, and quartiles have been calculated for distance, weight of packages, delivery rating, and delivery cost.

These measures offer you information about typical delivery conditions available and the degree of variability; also, the measures show one the range of values for operations.

5.1 Calculation Using Python

Descriptive statistics for key numerical variables

```
numeric_columns = [  
    "distance_km",  
    "package_weight_kg",  
    "delivery_rating",  
    "delivery_cost"  
]  
  
descriptive_stats = df_eda[numeric_columns].describe()  
  
print(descriptive_stats)  
  
To specifically compare the mean and median:  
central_tendency = pd.DataFrame({  
    "Mean": df_eda[numeric_columns].mean(),  
    "Median": df_eda[numeric_columns].median(),  
    "Standard Deviation": df_eda[numeric_columns].std()
```

```
})
```

```
print(central_tendency)
```

5.2 Actual Results

Variable	Mean	Median	Standard Deviation	Minimum	Maximum
Distance (km)	150.39	151.00	86.41	3.60	297.10
Package Weight (kg)	25.15	25.15	14.37	0.67	49.52
Delivery Rating	3.67	4.00	1.15	1.00	5.00
Delivery Cost	864.94	867.54	435.71	95.67	1,632.72

Table 3

5.3 Analysis of Delivery Distance

On average, deliveries traveled 150.39 kilometers, which is very similar to the median distance of 151.00 kilometers.

Since both the mean and the median values are almost equal in this case, it can be concluded that in general the distance that the deliveries traveled is not exaggerated error-wise.

The variability of destinations measured by the standard deviation of 86.41 km indicates a wide variety of distances. Indeed, trips ranged from 3.6 km to 297.1 km, which means that deliveries go through both short and long distances.

Shipping significance

Due to the large variability of the distance traveled, the requirements in terms of logistics would also differ from one shipment to another. Thus, distance is an important criterion for assessing the cost and efficiency of delivery.

5.4 Analysis of Package Weight

The average weight of a package is 25.15 kg, while the median weight of the same package is also 25.15 kg.

In terms of standard deviation, it comes out to be 14.37 kg having enormous variation in the weights of the shipments. The weight of the lightest package would be around 0.67 kg and the heaviest package would weigh about 49.52 kg.

Logistics implication

The weight of a package could influence the selection of vehicles, handling requirements, transportation capacity, and costs of delivery. The huge variation in weight suggests that the logistics operation needs to account for very different requirements.

5.5 Analysis of Delivery Rating

On average, the delivery rating was 3.67 out of 5 with 4 being the median rating.

Rating values range from 1 to 5 with the standard deviation being around 1.15.

It can be said that since mean and median are unequal, therefore the ratings' distribution cannot be characterized as symmetric. The mean being lower than 4 means that there were enough low ratings that brought the overall average down.

Practical implications

Customer rating can serve as an important performance indicator. Further research can check if low ratings correlate with the rate of delivery or specific modes of delivery used, as well as weather conditions or delivery providers.

5.6 Analysis of Delivery Cost

The mean cost of delivery is recorded to be 864.94 while median cost value is evaluated to be equal to 867.54.

The standard deviation has the value of 435.71 indicating high level of dispersion of delivery costs.

The lowest delivery charge is about 95.67 and the highest charge is about 1,632.72.

The comparatively big standard deviation indicates that there is high variability of delivery costs throughout the transactions that are being examined.

Logistics implications

It is necessary to know the causes of this variability to get good command of the logistics expenses. Distance and weight of the parcel will be studied in the order of correlation and visualization analysis.

5.7 Overall Interpretation

The descriptive data highlights a few initial observations:

- On average, the typical delivered distance is about 150 km.
- Standard weight for packages stands at roughly 25 kg.
- Customer satisfaction is around at 3.67/5.
- Average cost for delivering packages is estimated at 864.94.
- Delivery distance and cost display significant variability.
- Standard weights of packages also seem to vary a lot.
- Customer satisfaction could be regarded as moderate.

Nonetheless, descriptive statistics alone can't explain how all these variables are interconnected and what leads to variability among them.

5.8 Why These Statistics Matter

Statistic	Purpose in Logistics Analysis
Mean	Identifies the typical operational value
Median	Shows the central observation and helps assess distribution
Standard deviation	Measures operational variability
Minimum	Identifies the lowest observed value
Maximum	Identifies the highest observed value
Quartiles	Shows how observations are distributed across the dataset

Table 4

CONCLUSION

Descriptive will assist in finding baseline conditions in logistics operations. The most interesting information presented here is the discrepancy in delivery distance and delivery cost, which implies that these items can be subject to further observations.

6. Distribution Analysis and Visualisations

Descriptive statistics give quantitative summaries but fail to describe the distribution of variables. That is why visual methods were used for the analysis of the distribution of the main quantitative logistics variables.

Four legal variables were considered:

- distance in kilometers
- weight of the package in kg
- delivery rating
- the cost of delivery

Two main visual techniques were implemented:

- Histogram — for the assessment of the distribution and concentration of observations.
- Box plots — for determining spread, median, and any outliers.

6.1 Distribution of Delivery Distance

Python Code

```
plt.figure(figsize=(8, 5))
```

```
sns.histplot(  
    df_eda["distance_km"],  
    bins=30,  
    kde=True  
)
```

```
plt.title("Distribution of Delivery Distance")
```

```
plt.xlabel("Distance (km)")  
plt.ylabel("Number of Deliveries")  
plt.show()
```

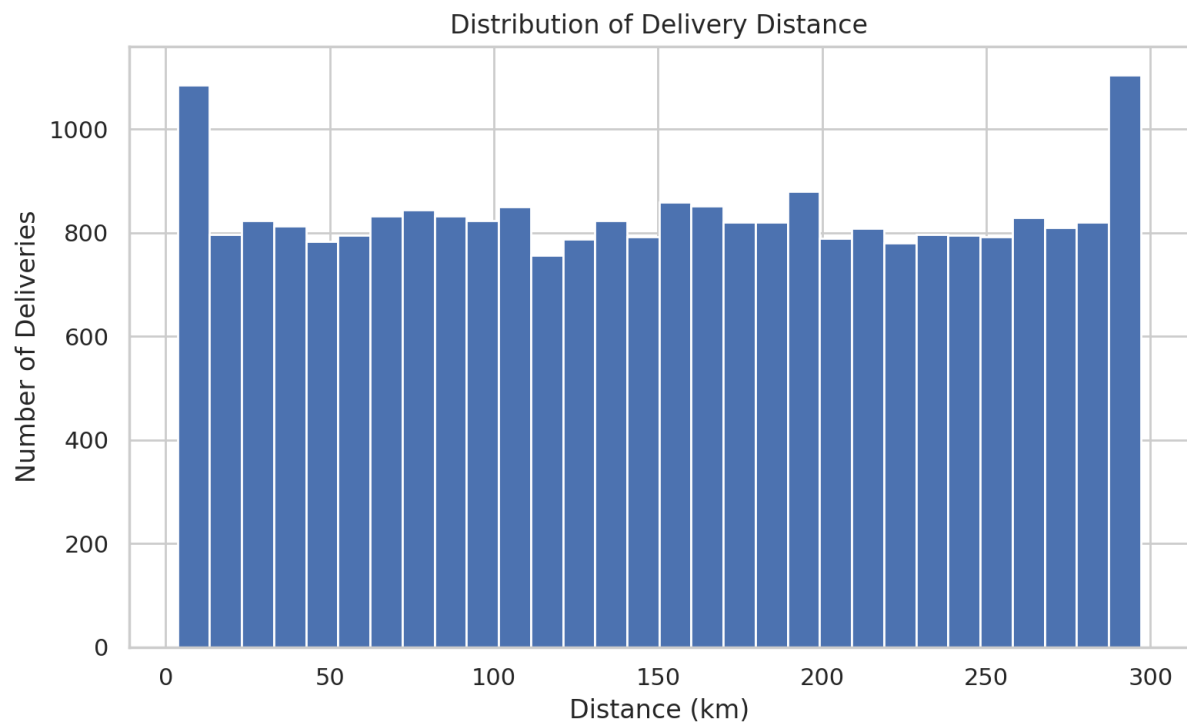


Fig1.

Interpretation

Distance data is fairly spread out and has a fairly symmetrical distribution shape.

The average delivery distance is of 150.39 km and the median distance is 151.00 km. Based on these values, we can say that distance data is centered around a distance of 150 km.

The distance data also has a 10th percentile of almost 30.6 km and a 90th percentile of approximately 270.4 km.

Why was this visualization selected?

A histogram is a good option to visualize distance as distance is a continuous variable and the histogram can show how deliveries are concentrated as well as the skewness of the distribution.

Logistics usefulness

The broad spread of distance means that the company has to deal with drastically different transportation requirements. Therefore, distance should be taken into account in cost consideration.

6.2 Distribution of Package Weight

Python Code

```
plt.figure(figsize=(8, 5))

sns.histplot(
    df_eda["package_weight_kg"],
    bins=30,
    kde=True
)

plt.title("Distribution of Package Weight")
plt.xlabel("Package Weight (kg)")
plt.ylabel("Number of Deliveries")
plt.show()
```

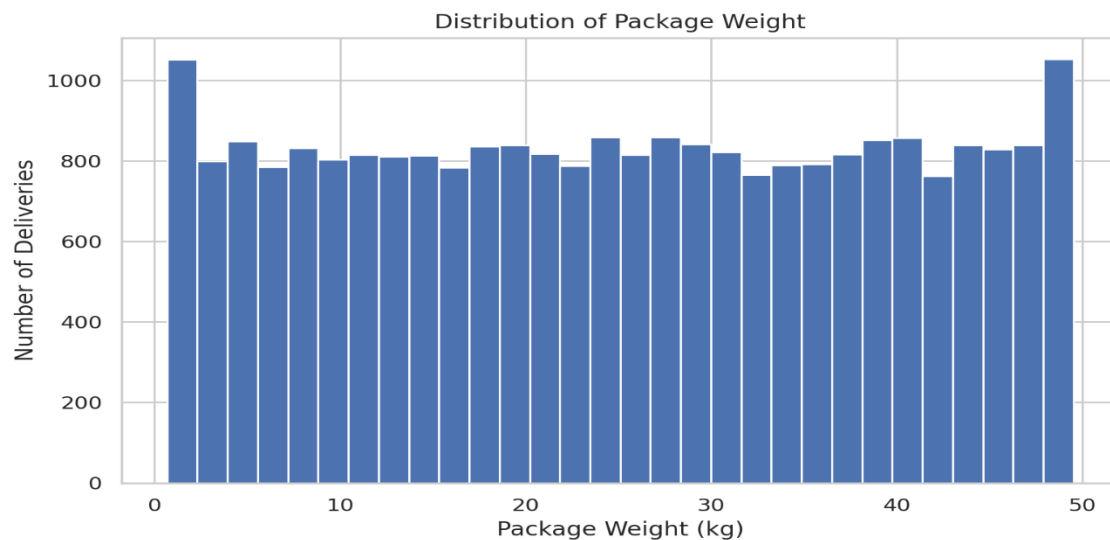


Fig2.

Explanation

The weights of packages are almost symmetrically distributed around the average.

In this case, the mean and median are around 25.15 kilograms, while the weights themselves range from 0.67 to 49.52 kilograms.

The values for the 25th and 75th percentiles are equal to 12.68 and 37.66 kg, correspondingly.

Reasons for selecting this method of visualization:

The use of a histogram is appropriate when there is a need to analyse the level of concentration and dispersion of weights of packages.

Logistics implications

The presence of large-scale weight variability indicates that there may be drastic differences in terms of vehicle capacity and conditions of delivery.

6.3 Distribution of Delivery Rating

Python Code

```
plt.figure(figsize=(8, 5))

sns.countplot(
    data=df_eda,
    x="delivery_rating"
)

plt.title("Distribution of Delivery Ratings")
plt.xlabel("Delivery Rating")
plt.ylabel("Number of Deliveries")
plt.show()
```



Fig3.

Interpretation

Delivery rating is quantifiable through a discrete scale of a five-point system, which justifies the use of a count plot, unlike other cases where a histogram would be helpful due to distance or weight concepts.

The median for ratings is 4, while their average stands at an approximate value of 3.67.

With the help of count plot that presents data about the total number of deliveries by rating, this graph indicates that the highest values dominate delivery rating dimensions. However, low ratings are still present in the evaluation process. This fact can be interpreted as a sign of customers' positive attitude toward delivery services.

Reason for selecting particular visualization

Based on the count plot presented, it makes it easy to quantify the overall performance of a delivery service instead of treating the rating as a continuous variable.

Logistics significance of findings

Customer ratings can be treated as an operational performance indicator and verified by further research as to whether low ratings are connected to delivery delays, weather conditions or specific partners.

6.4 Distribution of Delivery Cost

Python Code

```
plt.figure(figsize=(8, 5))
```

```
sns.histplot(  
    df_eda["delivery_cost"],
```

```
bins=30,  
kde=True  
)  
  
plt.title("Distribution of Delivery Cost")  
plt.xlabel("Delivery Cost")  
plt.ylabel("Number of Deliveries")  
plt.show()
```

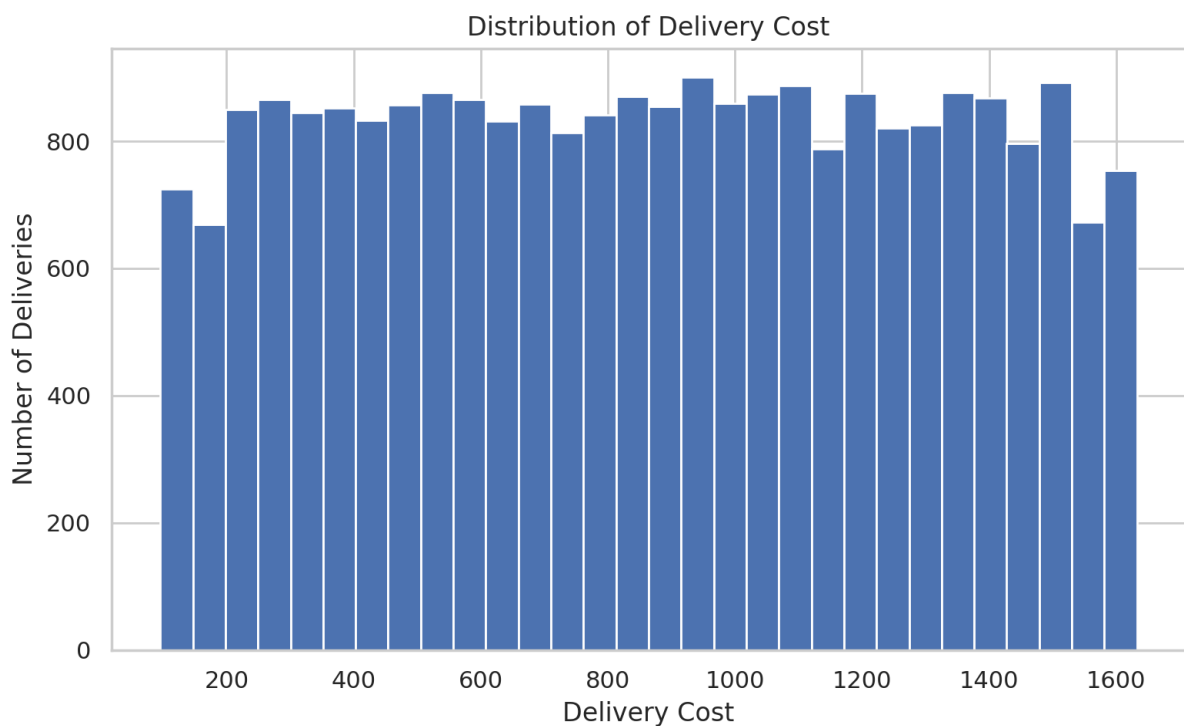


Fig4.

Interpretation

Delivery cost is almost evenly distributed through the range observed in the study.

The average delivery cost is just about 864.94 in comparison with the median of 867.54 which also means that the costs are distributed almost equally around the central value which is between 865 and 868.

The lowest cost is around 95.67 while the highest observed is approximately 1,632.72.

Why this visualization was chosen?

A histogram allows the management to see how much delivery costs vary from each transaction.

Logistics implication

The extensive variance in the delivery cost proved that delivery costs needs to be analyzed so that the planning of logistics operations can be improved. The next step is to look into what functions cause the variance.

6.5 Box Plot Analysis

Box plots were also used to compare the spread of the numerical variables and visually identify potential extreme observations.

Because the variables use different units, each variable should be visualized separately rather than putting all four raw variables on the same axis.

Example Python Code

```
plt.figure(figsize=(8, 5))
```

```
sns.boxplot(  
    y=df_eda["delivery_cost"]
```

)

```
plt.title("Box Plot of Delivery Cost")
```

```
plt.ylabel("Delivery Cost")
```

```
plt.show()
```

The same approach can be applied to distance, package weight, and rating.

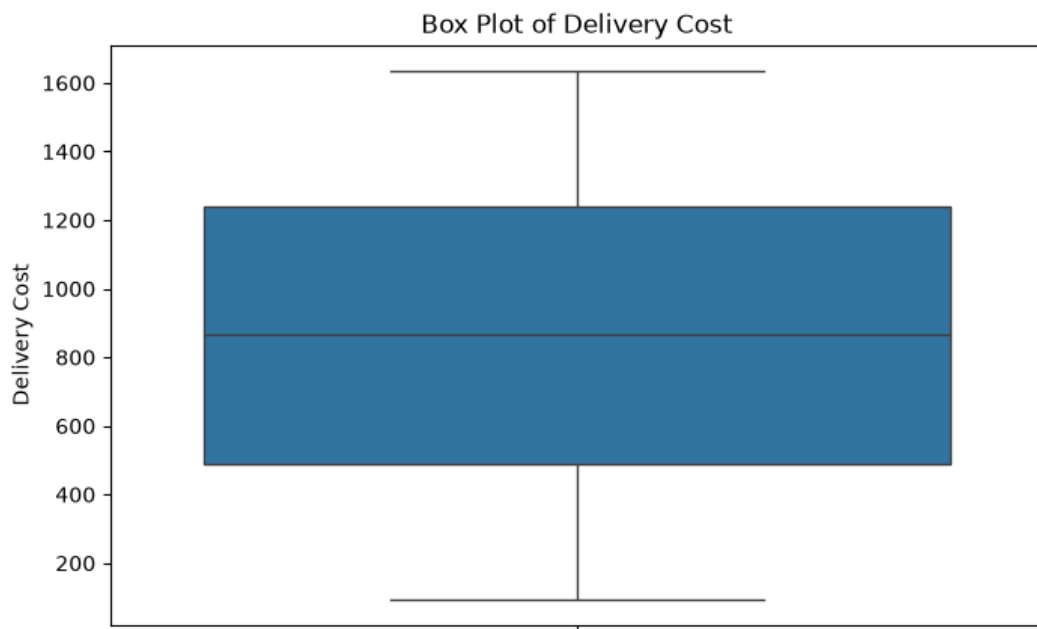


Fig5.

Interpretation

The box-plot analysis is consistent with the statistical IQR analysis performed earlier. **No IQR-based outliers were identified** in any of the four numerical variables.

Therefore, there is no statistical justification for removing extreme observations.

6.6 Distribution Findings

The visualization analysis provides the following observations:

Variable	Distribution Finding	Business Interpretation
Distance	Approximately symmetric	Deliveries cover a broad range of distances
Package weight	Approximately symmetric	Shipment weights vary considerably
Rating	Concentrated toward higher ratings	Overall satisfaction is moderately positive
Delivery cost	Approximately symmetric	Delivery expenditure varies substantially
Numerical outliers	None by IQR	No statistical outlier treatment require.

Table5

KEY INSIGHT

The distance, weight of the package, and cost of delivery distributions are evenly spread out because both their means and medians are found to be very similar.

This is beneficial as it shows that the key performance variables are not distorted greatly by a few extreme values.

However, the larger ranges imply rich operational variations that require investigation through relationship and categorical analysis.

6.7 Why Visualisations matter?

Visual representations give information that cannot be offered just through descriptive data.

For instance, learning that the mean distance of delivery is 150.39 km indicates the central metric, yet, the histogram proves that the deliveries are distributed in a much wider range.

The same applies to the mean rating of 3.67 that does not indicate even how ratings differ from 1 and 5. The count plot makes it visible at once.

Hence, key advantages of visual representations for decision-making in the logistics sector include such aspects as:

- availability of delivery activity distribution;
- availability of variation in measures related to the operational process;
- presence of satisfaction levels;
- potential occurrence of outliers;
- existence of need for further investigations.

CONCLUSION

The analysis of distribution has demonstrated that the variables of logistics numbers have pronounced but rather uniform variability without automation of extreme as per IQR method.

7. Categorical Logistics Performance Analysis

After examining the numerical distributions, the next stage of the EDA focuses on categorical variables. The objective is to identify whether delivery performance differs across **delivery modes, weather conditions, delivery partners, regions, and vehicle types**.

A key performance metric used in this section is the **delay rate**:

$$\text{Delay Rate} = \frac{\text{Number of Delayed Deliveries}}{\text{Total Deliveries}} \times 100$$

This metric provides a more meaningful comparison than simply counting delayed deliveries because the categories contain different numbers of observations.

7.1 Delay Rate by Delivery Mode

Python Code

```
delay_by_mode = (
    df_eda.groupby("delivery_mode")["delayed"]
    .apply(lambda x: (x == "yes").mean() * 100)
    .sort_values(ascending=False)
)

print(delay_by_mode)
```

Actual Results

Delivery Mode	Number of Deliveries	Delay Rate
Express	6,233	73.78%

Same Day	6,279	32.54%
Two Day	6,302	0.43%
Standard	6,186	0.00%

Table6

Visualization

```
plt.figure(figsize=(8, 5))
```

```
sns.barplot(
    x=delay_by_mode.index,
    y=delay_by_mode.values
)
```

```
plt.title("Delay Rate by Delivery Mode")
```

```
plt.xlabel("Delivery Mode")
```

```
plt.ylabel("Delay Rate (%)")
```

```
plt.xticks(rotation=20)
```

```
plt.tight_layout()
```

```
plt.show()
```

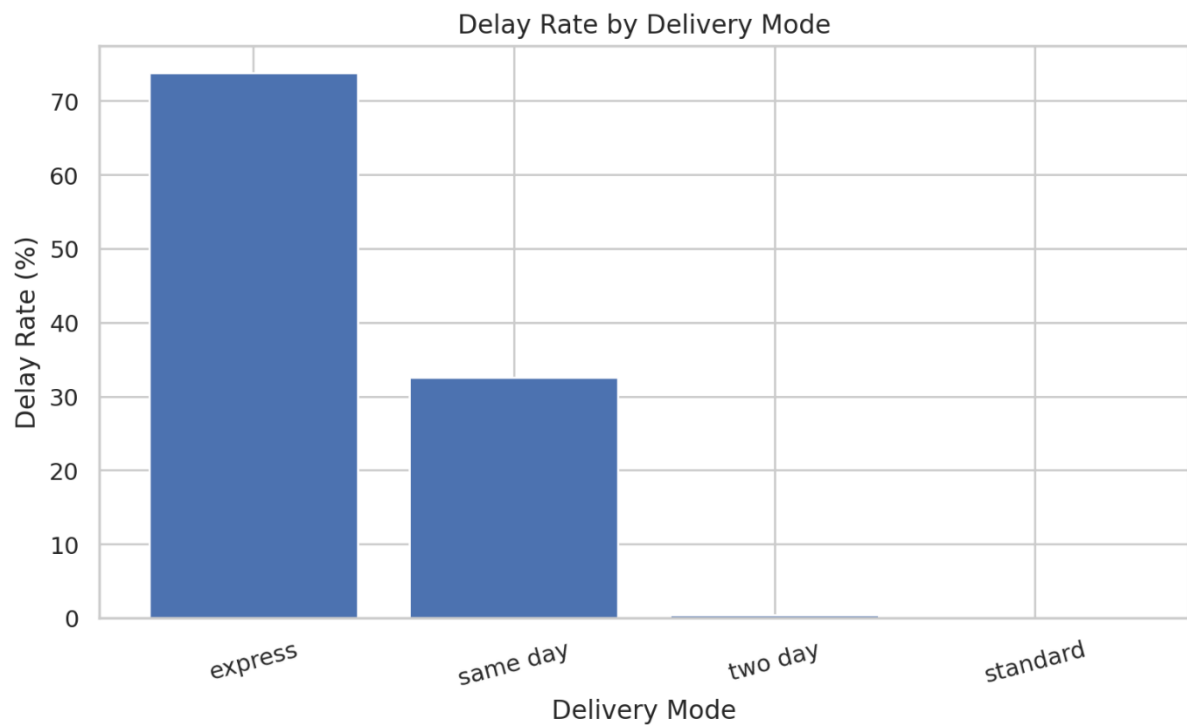


Fig6.

Interpretation

The delivery method is identified as the variable with the largest deviation in delay performance.

Express deliveries experience delays of 73.78%, which is significantly worse than the 32.54% rate of Same Day deliveries and only 0.43% for Two Day deliveries. Standard deliveries have an even better record of not having any delays.

Logistics implication

This is an important operational consequence. The very high degree of delays for Express deliveries may indicate that the daring promises of delivery are creating additional pressure on the operations.

The management has to look into:

- whether the Express orders require shorter delivery promises,
- whether Express deliveries are difficult to carry on certain routes,
- whether there is enough capacity during the busiest periods and
- whether there is proper classification of delays.

This should be regarded as a finding requiring further analysis rather than evidence that Express deliveries cause the delays.

The intentions behind the use of bar chart.

A bar chart is appropriate because delivery mode is a categorical variable and the objective is to compare the delay rate between categories.

7.2 Delay Rate by Weather Condition

Weather can affect transportation conditions and therefore potentially influence delivery performance.

Python Code

```
delay_by_weather = (  
    df_eda.groupby("weather_condition")["delayed"]  
    .apply(lambda x: (x == "yes").mean() * 100)  
    .sort_values(ascending=False)  
)
```

```
print(delay_by_weather)
```

Actual Results

Weather Condition	Delay Rate
Stormy	41.45%
Rainy	37.35%
Foggy	30.32%
Clear	17.43%
Hot	17.12%
Cold	16.02%

Table7.

Visualization

```
plt.figure(figsize=(8, 5))
```

```
sns.barplot(  
    x=delay_by_weather.index,  
    y=delay_by_weather.values  
)
```

```
plt.title("Delay Rate by Weather Condition")
```

```
plt.xlabel("Weather Condition")
```

```
plt.ylabel("Delay Rate (%)")
```

```
plt.xticks(rotation=20)
```

```
plt.tight_layout()
```

plt.show()

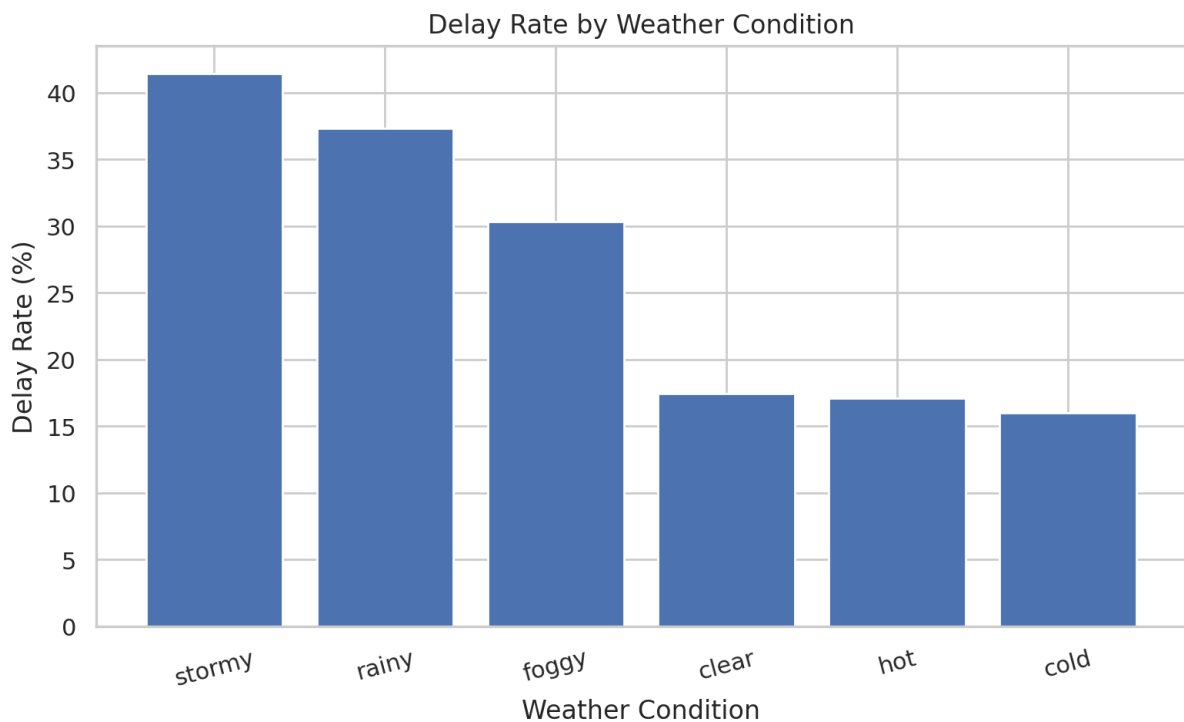


Fig7.

INTERPRETATION

There is consistent evidence that severe weather leads to more delays.

In fact, delays are most noticeable during stormy conditions (41.45%), followed by rainy (37.35%) and foggy (30.32%) conditions.

Cold weather had the least amount of delay at only 16.02%.

Implication for Logistics

The data shows that severe weather needs to be taken into account in logistics operations. Some strategies that logistics companies should consider:

- increase delivery buffers during severe weather,
- do route planning in advance,
- perform dispatching depending on the weather conditions, and

- communicate effectively with customers in anticipation of the bad weather.

Once again, this is an association, not causation.

Particulars of Bar Charts

Bar charts are perfect for visually illustrating the differences between weather types and can be used for the comparison of different groups.

7.3 Delay Rate by Delivery Partner

Delivery partners can also be compared to identify potential differences in operational performance.

Python Code

```
delay_by_partner = (  
    df_eda.groupby("delivery_partner")["delayed"]  
    .apply(lambda x: (x == "yes").mean() * 100)  
    .sort_values(ascending=False)  
)  
  
print(delay_by_partner)
```

Actual Results

Delivery Partner	Delay Rate
Xpressbees	28.27%
Ekart	27.42%

Shadowfax	27.30%
Amazon Logistics	27.22%
Blue Dart	26.95%
Ecom Express	26.63%
DHL	26.34%
FedEx	25.16%
Delhivery	

Table8.

Visualization

```
plt.figure(figsize=(10, 5))
```

```
sns.barplot(
    x=delay_by_partner.values,
    y=delay_by_partner.index
)
```

```
plt.title("Delay Rate by Delivery Partner")
```

```
plt.xlabel("Delay Rate (%)")
```

```
plt.ylabel("Delivery Partner")
```

```
plt.tight_layout()
```

```
plt.show()
```

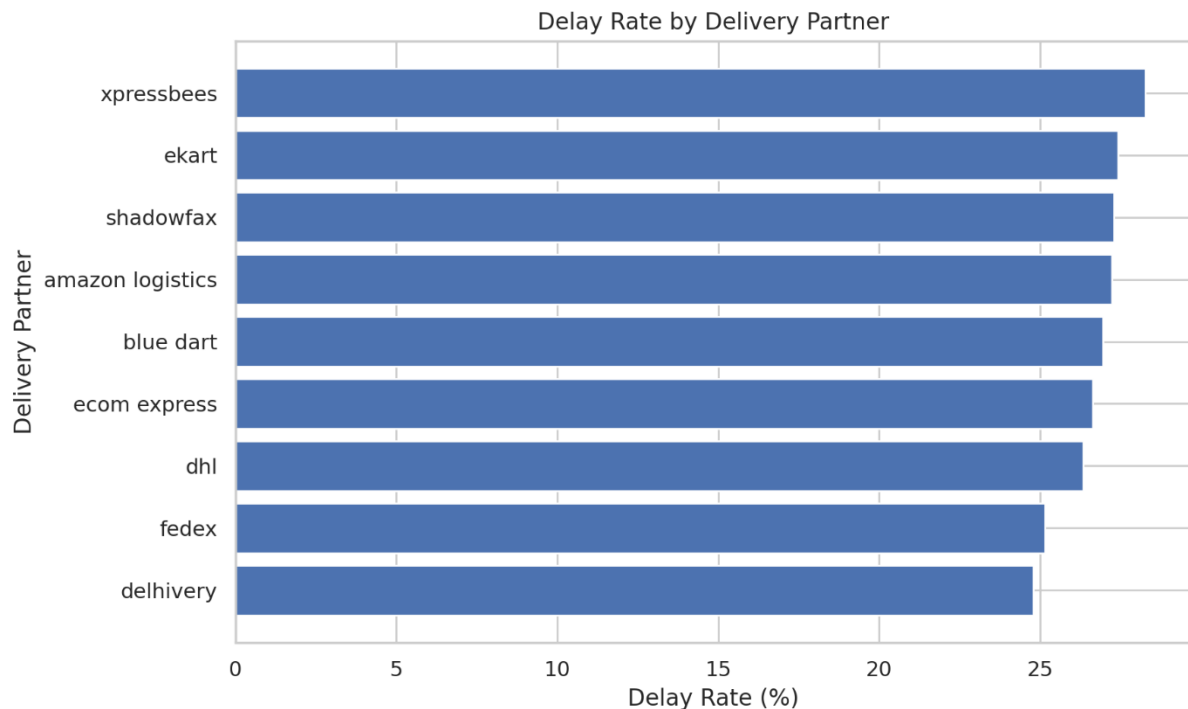


Fig8.

INTERPRETATION

When comparing partner delay rates, it should be noted that the gap between the maximum and minimum is smaller than the variation that exists between delivery methods.

In this case, Xpressbees has the highest percentage at 28.27% while Delhivery has the lowest rate at 24.80%.

Logistics Acumen

Given the narrow range of variability, delivery partner selection is unlikely to be responsible for most of the variance in delays. More likely, a combination of operational factors, such as delivery method and weather, accounts for the differences seen here.

Nevertheless, this approach of benchmarking at the partner level can still be useful in identifying potential areas for improvement.

Reasons for a horizontal bar chart

With the nine different delivery partners, a horizontal bar chart is the best choice as it facilitates reading of category names and straightforward comparison among partners.

7.4 Delay Rate by Region

Python Code

```
delay_by_region = (  
    df_eda.groupby("region")["delayed"]  
    .apply(lambda x: (x == "yes").mean() * 100)  
    .sort_values(ascending=False)  
)  
  
print(delay_by_region)
```

Actual Results

Region	Delay Rate
Central	27.25%
West	26.95%
South	26.78%
North	26.57%
East	25.80

Table9.

Visualization

```
plt.figure(figsize=(8, 5))

sns.barplot(
    x=delay_by_region.index,
    y=delay_by_region.values
)

plt.title("Delay Rate by Region")
plt.xlabel("Region")
plt.ylabel("Delay Rate (%)")
plt.tight_layout()
plt.show()
```

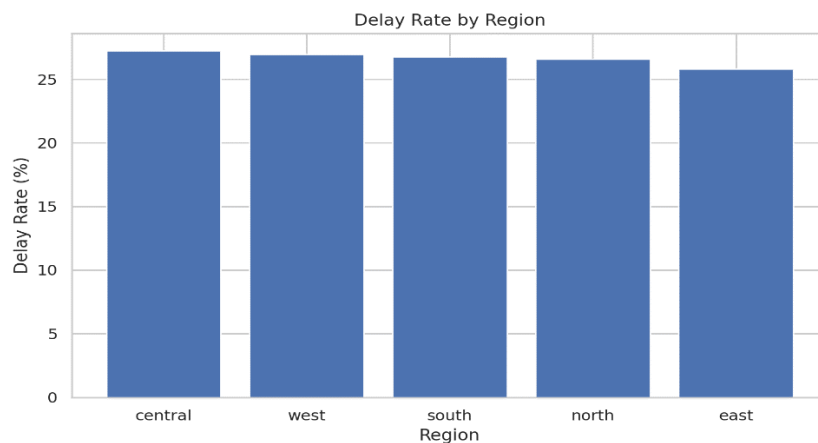


Fig.9

INTERPRETATION

Delivery delays in the various regions don't have a significantly different percentage because of the relatively close delay ratios which are in between 25.80% in the East region and 27.25% in the Central region.

Implication for Logistics

In the market, it is very recommended to analyze the regions along with other categories such as different delivery methods, climate conditions, vehicle types and distance.

7.5 Delay Rate by Vehicle Type

Python Code

```
delay_by_vehicle = (  
    df_eda.groupby("vehicle_type")["delayed"]  
    .apply(lambda x: (x == "yes").mean() * 100)  
    .sort_values(ascending=False)  
)  
  
print(delay_by_vehicle)
```

Actual Results

Vehicle Type	Delay Rate
Truck	27.04%
Bike	27.04%
Van	26.77%
EV Van	26.63%
EV Bike	26.43%
Scooter	26.14%

Table10.

INTERPRETATION

The delay rates for the vehicle types are very alike; they range from about 26.14% to 27.04%. Thus, the vehicle type does not display substantial differences in the overall delays within this dataset.

From a logistics perspective, this means that merely changing the type of vehicle may not really help to reduce delays significantly. The management should focus on researching other operational factors that may be more relevant.

7.6 Comparative Findings

The categorical analysis reveals an important distinction:

Factor	Observed Variation in Delay Rate
Delivery Mode	Very High
Weather	High
Delivery Partner	Moderate
Region	Low
Vehicle Type	Very Low

Table11.

Key Insight

Delivery mode is the strongest categorical differentiator of observed delay rates in this dataset.

The difference between Express (73.78%) and Standard (0%) deliveries is particularly large.

Weather is the second major differentiator, with Stormy conditions producing a 41.45% delay rate compared with 16.02% under Cold conditions.

By comparison, regional and vehicle-level differences are relatively small

7.7 Business Interpretation

These findings suggest that logistics managers should avoid relying on a single operational variable when investigating delays.

For example, a high Express delay rate could potentially be influenced by:

Express delivery requirements + route distance + weather + operational capacity

Therefore, the next stage of analysis should examine **relationships between numerical variables**, particularly the relationship between distance, package weight, delivery cost, and customer rating.

Conclusion

Categorical EDA identified substantial differences in delay performance across delivery modes and weather conditions, while partner, region, and vehicle differences were comparatively smaller.

8. Correlation Analysis

The analysis of the correlations was carried out between major numerical factors involved in logistics. It helps to provide an understanding of cost-generating factors and the extent to which the operational factors are correlated to each other.

The major variables examined are:

- distance_km
- package_weight_kg
- delivery_rating
- delivery_cost

8.1 Correlation Matrix

The Pearson correlation coefficient was used.

The correlation coefficient ranges from **-1 to +1**:

- **+1** → Perfect positive relationship
- **0** → No linear relationship
- **-1** → Perfect negative relationship

Python Code

```
numeric_columns = [  
    "distance_km",  
    "package_weight_kg",  
    "delivery_rating",  
    "delivery_cost"  
]  
  
correlation_matrix = df_eda[numeric_columns].corr()
```

```
print(correlation_matrix.round(3))
```

Actual Results

Variable	Distance	Weight	Rating	Cost
Distance	1.000	0.002	-0.133	0.991
Weight	0.002	1.000	0.007	0.099
Rating	-0.133	0.007	1.000	-0.161
Cost	0.991	0.099	-0.161	1.000

Table12.

8.2 Correlation Heatmap

A heatmap provides a visual representation of the correlation matrix.

Python Code

```
plt.figure(figsize=(8, 6))
```

```
sns.heatmap(  
    correlation_matrix,  
    annot=True,  
    cmap="coolwarm",  
    fmt=".2f",  
    center=0  
)
```

```
plt.title("Correlation Matrix of Logistics Variables")
```

```
plt.tight_layout()
```

```
plt.show()
```

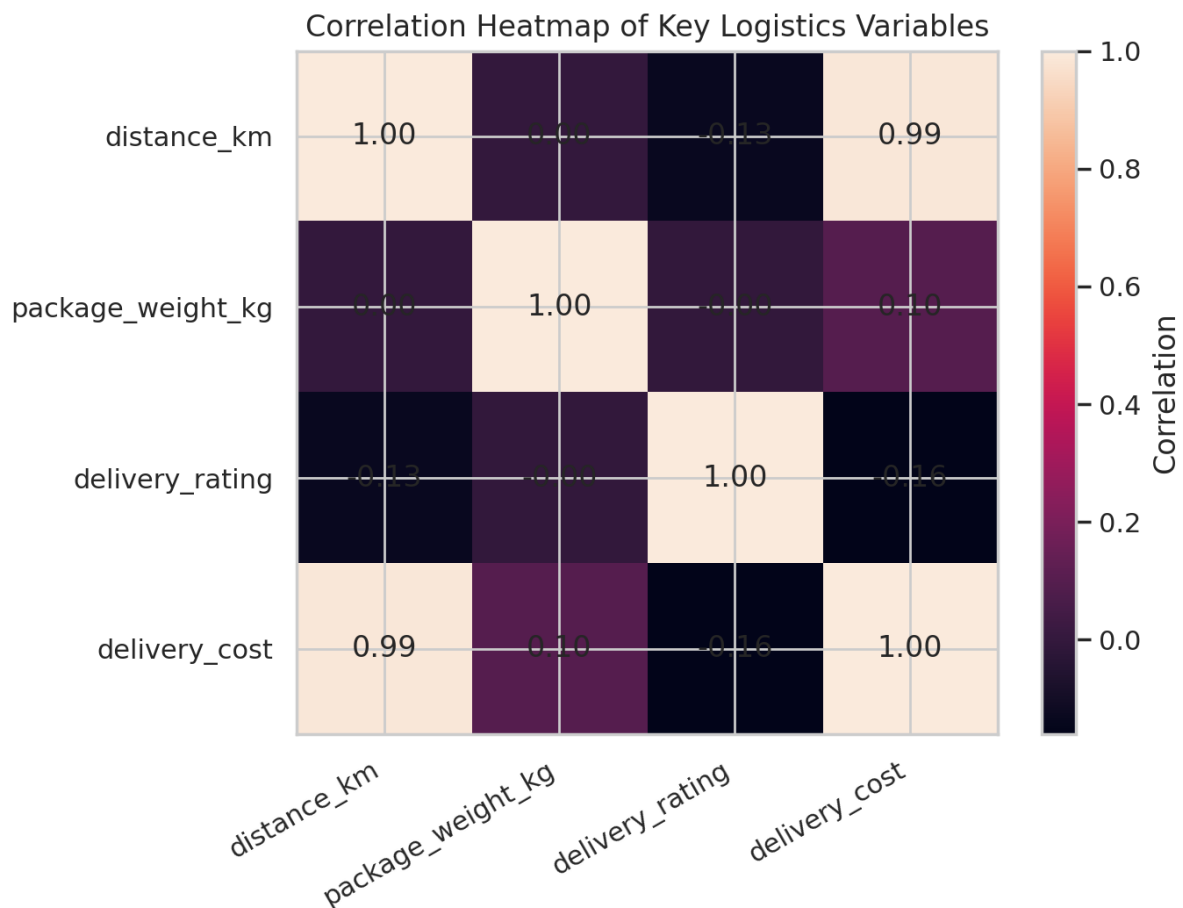


Fig10.

Why this visualization was selected?

A heatmap is appropriate because it allows several relationships to be examined simultaneously. Strong positive and negative relationships can be identified quickly without examining each correlation individually.

8.3 Distance and Delivery Cost

Within this data set, the highest correlation is given by

Distance and delivery expenses: $r = 0.991$

This illustrates very strong positive linear correlation.

This means that the two values are positively correlated in practice: the larger the distance travelled in connection with delivery, the more expensive delivery is likely to be.

Logistics implication

Distance is the most influential numerical cost factor in this data set.

This is crucial for logistics management as it indicates that variable distance in the provided route will cause high logistical expenses.

The management activities that could be implied from this piece of information would be:

- route optimizing,
- cutting unnecessary distance travelling,
- correct delivery zone planning,
- delivery consolidation when possible, and
- distance cost accounting.
- Important analytical note

The correlation of 0.991 does not prove causation on its own: it means strong linear correlation.

8.4 Package Weight and Delivery Cost

The link between the weight of the packages and how much it costs to deliver them is:

$r = 0.099$

This means there is a weak positive correlation.

While heavier packages do incur somewhat more costs, the correlation is less strong than the correlation between distance and costs.

Logistics conclusion

Weight measured alone fails to account for delivery cost fluctuations in this dataset.

Management must thus be careful to avoid conclusions implying that heavier packages account for the vast majority of the cost differences.

8.5 Distance and Package Weight

The relationship between distance and package weight is given by the following correlation:

$$r = 0.002$$

This correlation shows a negligible relationship in precise terms.

Interpretation

From the analysis made above, it could be interpreted that heavier packages do not tend to be transported over longer distances in this dataset.

This is important because it shows that distance and package weight provide fairly independent information.

Logistics implications

This means a logistics model may be able to utilize both pieces of information separately instead of assuming that one serves as a mere proxy to another.

8.6 Delivery Rating and Delivery Cost

The relationship between the rating of delivery and its cost can be expressed as follows:

$$r = -0.161$$

This shows that there is a weak negative correlation between the two variables.

High delivery costs seem to be related to lower customer ratings, although not strongly.

Logistics implications

Cost by itself cannot explain customer satisfaction. Other factors such as delays, mode of delivery, weather, and service experience can possibly be more relevant.

Hence, it is worth exploring this relationship together with the delay variable rather than separately.

8.7 Delivery Rating and Distance

The value of the correlation coefficient derived between the variables of distance and rating is:

$$r = -0.133$$

This means there exists a weak negative type of relationship.

Longer delivery distances are associated with marginally lower customer ratings, but this relation is insignificant.

Logistics interpretation

While long distances in some scenarios might affect customer satisfaction, it seems that distance on its own does not really answer for differences in ratings.

An interesting follow-up to this analysis would be to explore if the combination of being far and having a delayed shipment is a better predictor of satisfaction ratings.

8.8 Key Correlation Findings

Relationship	Correlation	Strength	Business Interpretation
Distance → Cost	0.991	Extremely strong positive	Distance is the dominant numerical cost driver
Weight → Cost	0.099	Weak positive	Limited relationship with cost
Distance → Rating	-0.133	Weak negative	Longer distances are slightly associated with lower ratings
Cost → Rating	-0.161	Weak negative	Higher-cost deliveries have slightly lower ratings
Distance → Weight	0.002	Negligible	Distance and weight are essentially independent
Weight → Rating	0.007	Negligible	Weight has almost no linear relationship with rating

Table13.

8.9 Main Analytical Insight

The correlation analysis showed that:

The distance of the shipment has a very strong positive correlation with the delivery charges, with the correlation coefficient r equal to 0.991; on the other hand, the weight of the package has only a weak correlation, with the coefficient equal to $r = 0.099$.

This means that the distance of the transportation should be one of the main factors affecting costs.

At the same time, the rather weak nature of the relationships in terms of customer rating signifies that customer satisfaction cannot be determined by quantitative indicators of logistics.

8.10 Restrictions of correlation analysis

In general, correlation analysis is helpful when analyzing linear relationships, but it has disadvantages.

A correlation:

- does not imply a causal relationship,
- only looks at linear relationships,
- does not take into account all categories,
- hides differences among groups, and
- is best viewed along with visual representation and business meaning.

Thus, results of correlation analysis will be interpreted along with results of categorical analysis and visualization.

CONCLUSION

Based on the results of the correlation analysis, distance was found to be the most significant numerical variable in determining the delivery cost, with a correlation of 0.991. While package weight is weakly associated with the cost, customer rating is negatively correlated to both distance and cost.

9. Relationship Visualisations and Analytical Interpretation

Thus far, the prior sections have shown that there is a strong link between the delivery distance and the delivery cost, while delivery method and the weather feature significant differences in delay rates.

In this section, visual representations are used to help convey these relationships better. Each representation is selected based on the logistics question that it intends to solve.

9.1 Distance vs. Delivery Cost

A scatter plot is used to examine the relationship between delivery distance and delivery cost.

Python Code

```
plt.figure(figsize=(9, 6))

sns.scatterplot(
    data=df_eda,
    x="distance_km",
    y="delivery_cost",
    alpha=0.5
)

plt.title("Relationship Between Delivery Distance and Delivery Cost")
plt.xlabel("Delivery Distance (km)")
plt.ylabel("Delivery Cost")
plt.tight_layout()
```

plt.show()

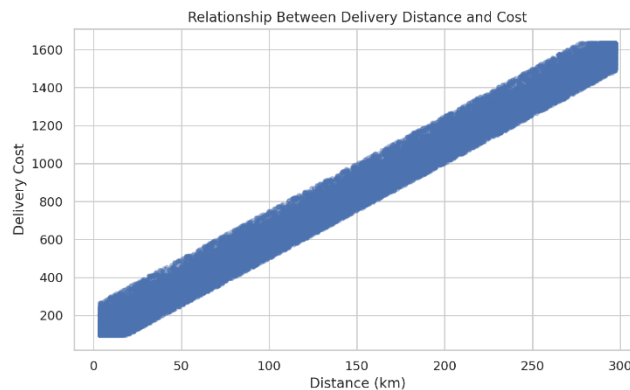


Fig11.

INTERPRETATION

The scatter plot ought to present evidence of a strong upward linear trend in accordance with the correlation coefficient of 0.991.

It indicates that as distance becomes longer, the cost of deliveries also experiences an upward trajectory.

Logistics implications

Distance should be given a due consideration as a very important factor in dealing with transportation costs.

This gives rise to actions such as the following:

- optimization of routes,
- delivery channel planning,
- elimination of unwanted miles,
- combination and consolidation of routes, and
- independent assessment of the costs of long-distance shipments.

Why scatter plot?

It's possible to use a scatter plot to show the mutual effect of distance and cost because these are continuous numerical variables, and it allows to visualize the nature of their relationship.

9.2 Package Weight vs. Delivery Cost

The next visualization examines whether heavier packages are associated with higher delivery costs.

Python Code

```
plt.figure(figsize=(9, 6))

sns.scatterplot(
    data=df_eda,
    x="package_weight_kg",
    y="delivery_cost",
    alpha=0.5
)

plt.title("Relationship Between Package Weight and Delivery Cost")
plt.xlabel("Package Weight (kg)")
plt.ylabel("Delivery Cost")
plt.tight_layout()
plt.show()
```

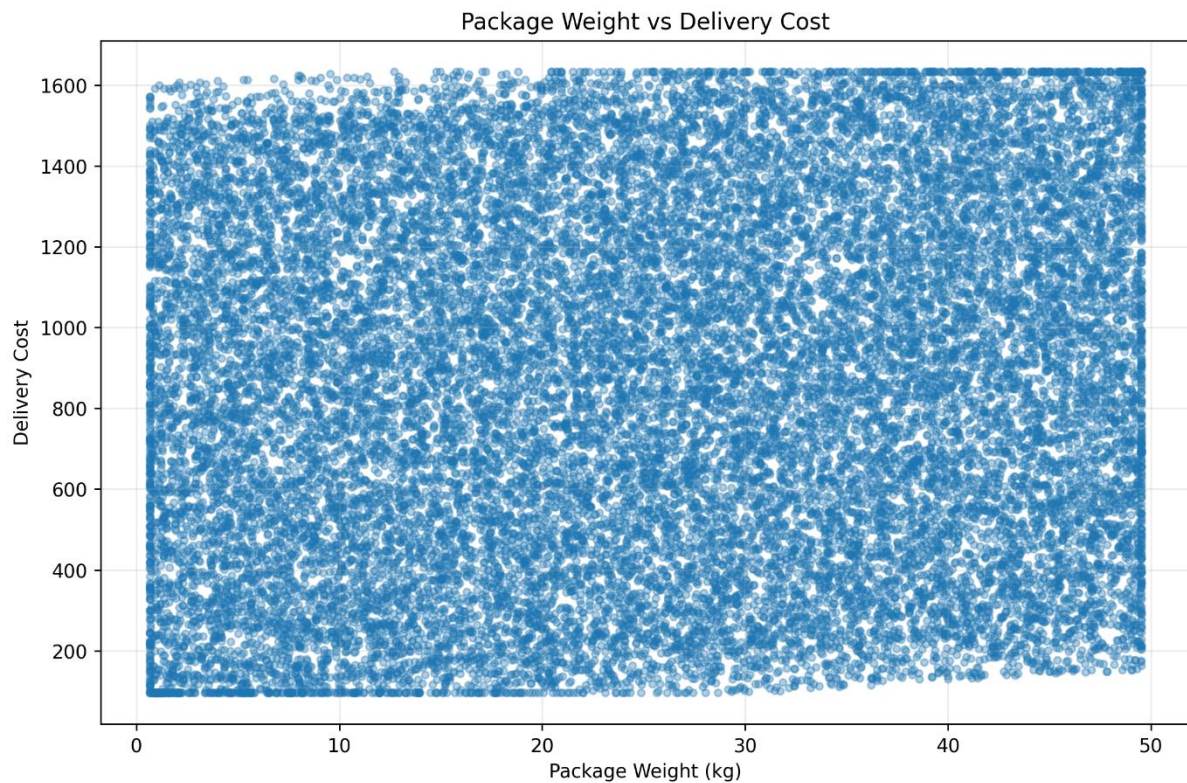


Fig12.

INTERPRETATION

The link between package weight and delivery costs is significantly less robust compared to the distance-cost link.

The correlation coefficient in this case is just 0.099, which shows that there is a weak positive linear relationship.

The scatter plot will therefore be much more spread out than the Distance vs. Cost scatter plot.

Logistics implications

Package weight may have an influence on delivery costs; however, it seems not to be the main numerical cost factor in this case.

Hence management should pay more attention to transportation distance in the process of designing cost-cutting options.

Why the scatter plot?

Since both variables are quantitative and continuous, a scatter plot is the best device to visualize their relationship.

9.3 Delivery Mode and Delay Rate

The delivery-mode analysis from Section 7 revealed substantial differences in delays.

A bar chart is used to communicate this finding clearly.

Python Code

```
plt.figure(figsize=(8, 5))
```

```
sns.barplot(  
    x=delay_by_mode.index,  
    y=delay_by_mode.values  
)
```

```
plt.title("Delay Rate by Delivery Mode")
```

```
plt.xlabel("Delivery Mode")
```

```
plt.ylabel("Delay Rate (%)")
```

```
plt.xticks(rotation=20)
```

```
plt.tight_layout()
```

```
plt.show()
```

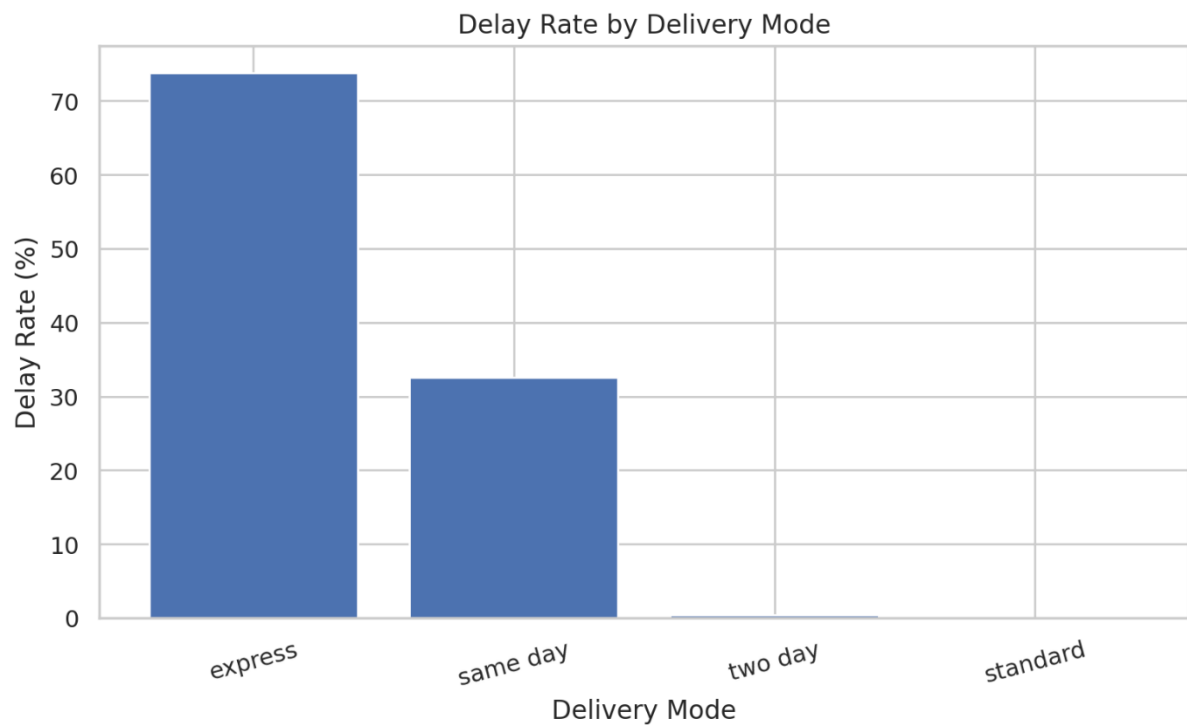


Fig13.

Main information

The delay rates measured are:

- Express: 73.78%
- Same Day: 32.54%
- Two Day: 0.43%
- Standard: 0.00%

Adequate interpretation

Based on the visualization the difference in the rates can be observed at once. Express deliveries have a significantly higher observed delay rate than in other delivery modes.

Logistics-related impact

This implies that it can be considered as a possible operational bottleneck which requires further research.

The management ought to investigate whether Express deliveries:

- have relatively strict promised delivery windows;
- involve more complicated routes;
- face limited capacity;
- are highly influenced by bad weather;
- use different operational practices.

It should not be assumed that only Express deliveries cause delays without further research.

Why to use a bar chart?

As the data on the delivery modes is categorical and the aim is to see the performance metric across the delivery modes, a bar chart will suit the best

9.4 Weather Condition and Delay Rate

Weather conditions were also analysed because transportation performance can potentially be affected by environmental conditions.

Python Code

```
plt.figure(figsize=(8, 5))
```

```
sns.barplot(
```

```
    x=delay_by_weather.index,
```

```
    y=delay_by_weather.values
```

)

```
plt.title("Delay Rate by Weather Condition")
```

```
plt.xlabel("Weather Condition")
```

```
plt.ylabel("Delay Rate (%)")
```

```
plt.xticks(rotation=20)
```

```
plt.tight_layout()
```

```
plt.show()
```

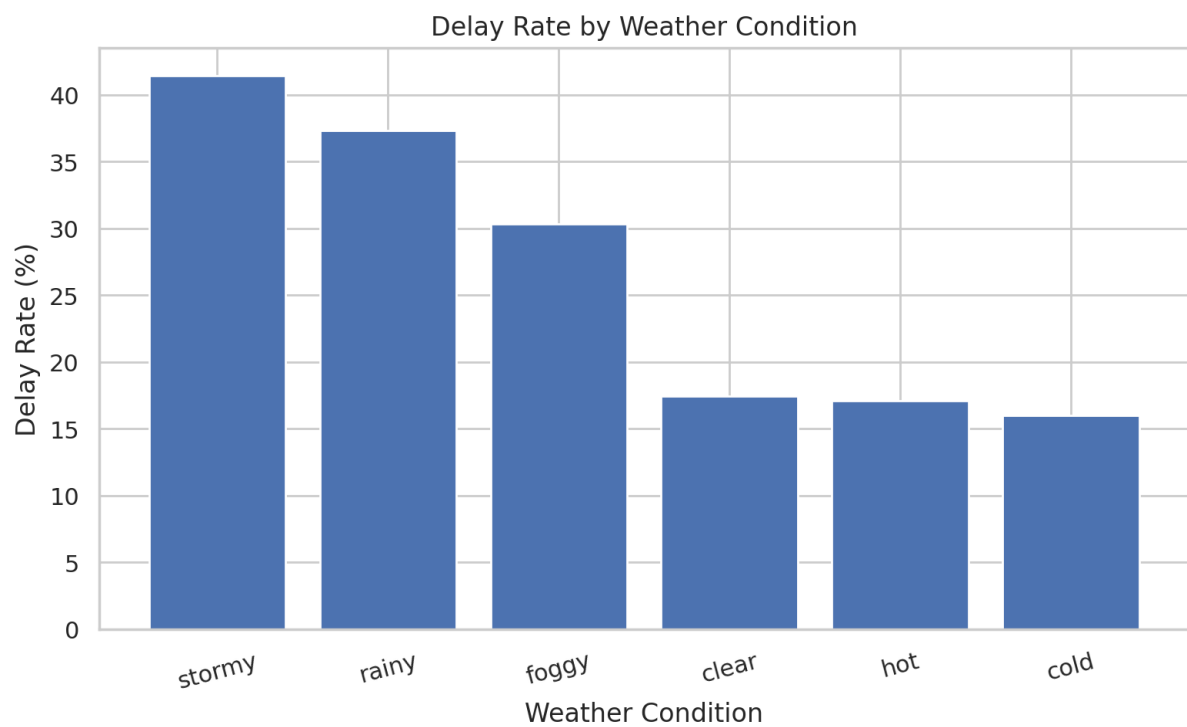


Fig14.

Key Finding

Weather	Delay Rate
---------	------------

Stormy	41.45%
Rainy	37.35%
Foggy	30.32%
Clear	17.43%
Hot	17.12%
Cold	16.02%

Table14.

Explanation

The flow chart illustrates the contrast between nasty and reasonably acceptable weather situations.

The maximum percentage related to delays is represented in the concept of unfavorable weather with 41.45%, followed by rainy weather with 37.35%.

Logistics implications

Taking weather into account could increase the reliability of delivery. When experiencing unfriendly weather, companies should think of:

- extra buffers for delivery,
- rerouting,
- early departure,
- alerting clients timely,
- watching the delivery closely.

Why a bar graph?

The weather influence belongs to the categorical variable category which makes the bar graph perfectly suitable for this comparison.

9.5 Correlation Heatmap

The final visualization combines the major numerical relationships into one graphic.

Python Code

```
plt.figure(figsize=(8, 6))
```

```
sns.heatmap(  
    correlation_matrix,  
    annot=True,  
    cmap="coolwarm",  
    fmt=".2f",  
    center=0  
)
```

```
plt.title("Correlation Heatmap of Logistics Variables")
```

```
plt.tight_layout()
```

```
plt.show()
```

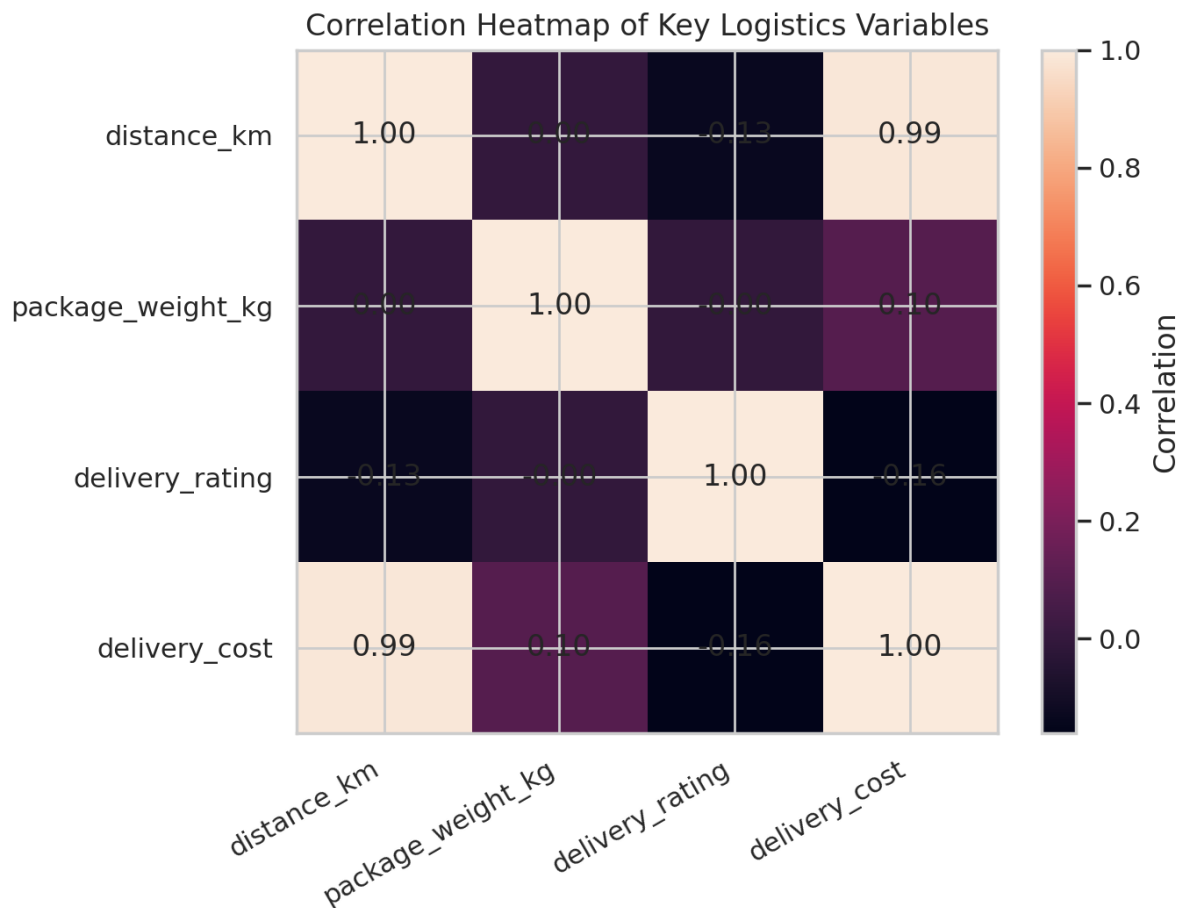


Fig15.

Interpretation

According to the heatmap, the strongest correlation in the dataset can be concluded.

Distance ↔ Delivery Cost = 0.991

to be specific, other data points are showing that

- Weight ↔ Cost = 0.099
- Distance ↔ Rating = -0.133
- Cost ↔ Rating = -0.161
- Distance ↔ Weight = 0.002
- Weight ↔ Rating = 0.007

Thus, the heatmap serves as a brief overview of the correlation between the numerical variables mentioned within EDA session.

The need of heatmaps?

Whenever few numerical variables should be assessed at the same moment, the heatmap would be especially helpful.

9.6 Visualization Summary

Visualization	Main Question	Key Finding
Distance vs Cost	Does distance relate to cost?	Very strong positive relationship
Weight vs Cost	Does weight relate to cost?	Weak positive relationship
Delay Rate by Mode	Which mode has the most delays?	Express: 73.78%
Delay Rate by Weather	Does delay rate differ by weather?	Stormy: 41.45%
Correlation Heatmap	How are numerical variables related?	Distance-cost correlation = 0.991

Table15.

9.7 Overall Summary of Information Visualizations

The visual information highlights three core observations:

1. Distance is one of the main factors influencing transportation costs.

The very solid distance versus cost relationship points out that distance traveled is closely correlated with transportation cost.

2. Express deliveries need to be monitored

The 73.78% delay rate in Express deliveries is significantly higher than the delay rates for other examples of the transportation modes, and it is also the biggest area of inconsistency among delivery types.

3. Weather is an important operational factor

There are much higher observed delays during stormy or wet weather conditions than in all of the other weather conditions.

Thus, these observations suggest that logistics should be assessed as a combination of cost, level of quality of service, mode of delivery, and weather conditions rather than as a single measure of transport and logistics performance.

Conclusion

The analysis of visualizations helps to articulate and decode the most significant trends presented in the data. The severest quantitative relation occurs between distance and cost of delivery. In contrast, major categorical differences are seen with respect to various delivery modes and weather situations.

10. Key Analytical Insights, Operational Bottlenecks and Cost Drivers

The EDA performed in the previous sections identified several patterns that are relevant to logistics performance. This section consolidates the numerical and categorical findings into actionable business insights.

10.1 Overall Delivery Performance

The overall observed delay rate in the dataset is **26.68%**.

The delivery-status distribution is:

Delivery Status	Percentage
Delivered	73.32%
Delayed	21.36%
Failed	5.31%

Table16.

This indicates that although the majority of deliveries were successfully delivered, delays and failures together represent a meaningful portion of the operational activity.

Business implication

A logistics organization should not focus only on successful deliveries. Monitoring the **26.68% overall delay rate** can help identify opportunities to improve service reliability.

10.2 Delivery Mode as a Major Bottleneck

Delivery mode is the most prominent categorical factor identified in the analysis.

Delivery Mode	Avg. Cost	Avg. Rating	Delay Rate
Express	880.43	2.72	73.78%
Same Day	929.87	3.54	32.54%
Two Day	829.71	4.18	0.43%
Standard	819.34	4.22	0.00%

Table17.

Key finding

Express deliveries show a delay of 73.78%, whereas Two Day and Standard deliveries show only 0.43% and 0%, respectively.

Average customer satisfaction ratings of Express deliveries are 2.72, while ratings for Standard deliveries stand at 4.22.

Interpretation

It is an essential combination:

Express → High delay rate → Lower customer satisfaction rating

Nonetheless, the analysis does not support the idea that Express delivery leads to low ratings directly. Various other factors might affect this relationship.

Recommendation

The management shall analyze the workflow of the Express service thoroughly. The investigation must include:

- the time windows promised for delivery,
- complexity of routes taken,
- possibility of dispatching to meet the volume,

- scale of orders affected,
- weather impact,
- distribution of resources, and
- understanding of the meaning of "delay" within the operational system.

10.3 Distance as the Primary Numerical Cost Driver

The correlation analysis identified:

Distance vs. Delivery Cost = 0.991

This is an exceptionally strong positive relationship.

The average delivery distance is **150.39 km**, while average delivery cost is **864.94**.

Interpretation

The strong relationship indicates that deliveries covering greater distances tend to have higher delivery costs.

Interestingly, average distances across delivery modes are quite similar:

Delivery Mode	Average Distance (km)
Same Day	151.16
Express	150.95
Two Day	150.81
Standard	148.61

Table18.

This suggests that the differences in average cost between delivery modes cannot simply be explained by large differences in average distance.

Recommendation

Route optimization should therefore be an important component of cost-management strategy. Potential approaches include:

- route consolidation,
- reducing unnecessary kilometres,
- optimized delivery-zone allocation,
- improved route sequencing, and
- monitoring cost per kilometre.

10.4 Weather as an Operational Risk Factor

Weather conditions show substantial differences in observed delay rates.

Weather	Delay Rate
Stormy	41.45%
Rainy	37.35%
Foggy	30.32%
Clear	17.43%
Hot	17.12%
Cold	16.02%

Table19.

Key finding

The delay rate in Stormy conditions is more than two times that in Cold conditions.

Interpretation

Bad weather is associated with longer delivery delays.

Recommendation

Weather information can be integrated in the operational strategy through:

- weather-based delivery buffers,
- anticipatory route planning,
- early dispatch during bad weather,
- real-time monitoring, and
- active communication with customers.

10.5 Delivery Partners Show Relatively Small Differences

Again, it should be kept in mind that these findings represent only associations and not causal evidence.

According to the analysis done regarding the delivery-partners, it can be observed that there exists only a small variation in the delay rates.

The delivery-partner with the highest rate of delay is:

Xpressbees having a rate of delay of 28.27%

The partner with the least amount of delays is:

Delhivery having a delay rate of 24.80%

The difference between the two partners mentioned can be calculated as:

3.47%.

Interpretation

Different delivery modes and weather conditions show significant divergence in delay rates while there is only minor variation between partners in terms of their respective delivery delays.

Recommendation:

Management must ensure that continued monitoring of the partners' performance is done without concluding that changing the delivery partners will bring positive contributions to the overarching issue of delays.

Furthermore, the performance of the partners must be analyzed in conjunction with:

- the mode of delivery adopted,
- distance,
- weather,
- shipment characteristics, and
- regional workload.

10.6 Region and Vehicle Type Are Less Differentiated

The delay rates across regions are as follows:

- Central Region: 27.25 percent
- Eastern Region: 25.80 percent
- The difference is a mere 1.45 percentage points.

The delay rates of different modes of transport are also close, as the scooter recorded a delay rate of 26.14 percent whereas trucks and bicycles account for a delay rate of 27.04 percent.

Interpretation

Both the different region and mode of transport cause similar but slight differences in the overall delay rate.

Business implications

Major operations should not be designed on the basis of these two variables alone. They are better utilized in the framework of a more complex model of logistics performance.

10.7 Key Bottlenecks and Cost Drivers

The EDA allows the major findings to be ranked:

Priority	Finding	Evidence	Business Significance
1	Express delivery delays	73.78% delay rate	Major service bottleneck
2	Distance-cost relationship	r = 0.991	Major numerical cost driver
3	Weather-related delays	41.45% stormy delay rate	Operational risk
4	Same-Day cost	929.87 average	Highest average delivery cost
5	Partner differences	24.80–28.27% delay	Moderate variation
6	Regional differences	25.80–27.25% delay	Limited variation
7	Vehicle differences	26.14–27.04% delay	Very limited variation

Table20.

10.8 Strategic Recommendations

Based on the comprehensive EDA findings, the following suggestions are given:

1. Examine the Express Delivery Operations

The Express delay rate, which is 73.78%, should be checked as it is the top operational concern.

2. Enhance the Route Management

Because the correlation between distance and costs is 0.991, reducing unnecessary travels should be the main cost-reduction line.

3. Create the Weather-Based Planning System

During rainy, stormy, and foggy weather, the delay rates significantly increase. Weather-related risks should be given proper consideration during delivery planning.

4. Control the Customer Satisfaction Level

The average rating for express deliveries equals to 2.72, which is much lower than the 4.22 rating of the standard deliveries. Therefore, it is essential to check the customer ratings regarding the express deliveries and the delay rate.

5. Do Not Overstate the Influence of Vehicles and Regions

Due to the small gap between vehicles and regions, one cannot consider these parameters as the main reasons for delays without additional proof.

6. Creation of a Multi-Factor Logistics Dashboard.

An effective operational dashboard might incorporate the following elements:

- Delay metrics
- Delivery costs
- Per kilometre costs
- Customer client rating
- Delivery methodology
- Weather conditions
- Logistics provider
- Geographical information

This would enable the management team to analyze cost-effectiveness and efficiency of service provision.

10.9 Overall Business Insight

The key takeaway from Project 3 is that the factors that determine logistics performance vary in relation to different outcomes.

- Distance → closely linked to costs
- whereas
- Delivery method + weather condition → closely linked to delay and
- Delay/state of service → may be connected to customer satisfaction.

Hence, they should stay away from using a single KPI in order to assess logistics performance.

CONCLUSION

The EDA highlighted specific areas that require the attention of the management. The calculation that showed the most significant numerical result is the distance-cost correlation of 0.991, while the most meaningful categorical outcome was the 73.78 percent share of delayed Express shipments. In addition, bad weather conditions prove to result in much higher delays.

The above results serve as firm evidence for the suggestions made earlier and show how EDA in Python makes data-driven decisions in logistics possible.

11. Recommendations and Strategic Action Plan

Based on the findings from the exploratory data analysis, a practical action plan can be developed to improve delivery reliability, control logistics costs, and strengthen customer satisfaction.

11.1 Priority-Based Recommendations

Priority	Area	Evidence from Dataset	Recommended Action
High	Express deliveries	73.78% delay rate	Review Express delivery capacity, promised time windows, and routing
High	Transportation cost	Distance–cost correlation 0.991	Implement route optimization and monitor cost per km
High	Adverse weather	Stormy delay rate 41.45%	Introduce weather-based planning and delivery buffers
Medium	Customer satisfaction	Express rating 2.72	Investigate relationship between delays and ratings
Medium	Delivery partners	Delay rates 24.80–28.27%	Benchmark partners using standardized KPIs
Low	Region	Delay rates 25.80–27.25%	Continue monitoring rather than making major changes
Low	Vehicle type	Delay rates 26.14–27.04%	Monitor performance but avoid unnecessary fleet changes

Table21.

11.2 Strategic Action plan

Step One - Initial Actions

First, it is necessary to assess the efficiency of Express delivery.

Its remarkable level of Express delay should become the first matter to assess.

Management should analyze:

- the number of orders sent via Express,
- the pledged delivery time,
- the real operational capacity,
- the particulars of the route,
- the weather conditions, and
- the availability of drivers/ means of transport.

Second, it is advisable to keep an eye on costs associated with the distance.

Taking into account that distance is strongly correlated with delivery cost, management should determine the appropriate KPI:

Delivery Cost per Kilometer

This criterion will help to detect the routes with too high transportation costs.

Phase 2 of Initial Steps

3. Involve weather-aware delivery strategizing

Storms, rain and fog cause high delays in delivery operations. The operating team can use the weather information for.

4. Always keep customer satisfaction in mind along with delays

Customer rating can't be looked at in isolation from the performance in operations.

A dashboard could be created and include comparisons: Delay Rates and Customer Ratings, and Delivery Modes

Stage 3 — Advancements over the Intermediate Term

5. Create Standards for Every Partner

Each partner must be assessed with the help of definite indicators such as:

- delays ratio,
- cost of delivery,
- feedback from clients,
- percentage of failures, and
- completed deliveries.

The minimal gap in the final delay ratios of various partners proves that benchmarking needs to be coupled with other factors.

6. Provide a Smart Logistics Report

A dashboard needs to allow for overseeing the processes in a real-time regime in terms of:

- the total number of deliveries,
- the number of delayed deliveries,
- the number of deliveries that were unsuccessful,
- delays ratio,
- average costs,
- average distance,
- feedback from the recipient,
- transport method,
- weather conditions and
- a partner that performed the delivery.

11.3 Strategic Roadmap

The overall strategy can be summarized as:



11.4 Expected business benefits

Proper use of these actions can help an organization:

- cut down costs without the need for transportation
- make Express deliveries more reliable
- prepare for adverse weather conditions more effectively
- enhance customer experience
- analyze ineffective routes
- evaluate delivery partnerships objectively
- support operational decisions with firm evidence of outcomes.

It's important to note that EDA points out relationships and patterns, as opposed to guaranteeing results. The actual outcome of the suggestions proposed needs to be assessed based on operations and performance data.

12. Conclusion

The purpose of this project was to use Exploratory Data Analysis to examine operational characteristics and performance trends of the Delivery_Logistics dataset containing 25,000 records.

In particular, descriptive statistics, distribution analysis, categorical analysis, correlation analysis and Python visualizations were used.

The findings include three major outcomes:

- A strong positive correlation of 0.991 between the distance and delivery cost shows that distance is a very significant factor in calculating costs.
- Express orders have been found to have an observed delay rate of 73.78%, which makes Express operations the biggest categorical bottleneck identified.
- Bad weather conditions lead to delays too, especially Stormy (41.45%) and Rainy (37.35%).

The other outcome of the investigation is that the situations may differ a bit depending on delivery partners, regions and vehicle types.

In conclusion, the project illustrates the potential of EDA in deriving operational insights from raw logistics data. By means of this analysis, managers do not have to make assumptions about priorities like route optimization, Express service investigation, weather influenced planning, and customer service improvement anymore.

13. References

1. Pandas Development Team, Pandas Documentation; this library is utilized for reading and working with data, description analysis, grouping, and correlation analysis.
2. Matplotlib Development Team, Matplotlib Documentation; this is a framework used for plotting and representation.
3. Seaborn Development Team, Seaborn Documentation; used for plotting different statistical graphs like histogram, bar graph, scatter plot, box plot, and heatmap.
4. Scikit-learn Developers, Scikit-learn Documentation; this library is used as a supporting reference for incorporating Python techniques of data analysis and preprocessing.
5. Python Software Foundation, Python Documentation; this library is utilized as a reference in the implementation of programming.
6. Delivery_Logistics DataSet, Delivery_Logistics(1).csv. Used as the main data set for analyzing and visualizing in Project 3.

14. Appendix A — Complete Python Code

```
#
=====
=====

# PROJECT 3
# EXPLORATORY DATA ANALYSIS AND VISUALIZATION
# Dataset: Delivery_Logistics
#
=====
=====

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# -----
# 1. LOAD DATASET
# -----

df = pd.read_csv("Delivery_Logistics(1).csv")

print("Original dataset shape:", df.shape)

# Create analytical copy
```

```
df_eda = df.copy()

# Treat delivery ID as an identifier
df_eda["delivery_id"] = df_eda["delivery_id"].astype(str)

# Remove non-informative variables identified in Project 2
df_eda = df_eda.drop(
    columns=[
        "delivery_time_hours",
        "expected_time_hours"
    ]
)

print("EDA dataset shape:", df_eda.shape)


# -----
# 2. BASIC DATA INFORMATION
# -----

print("\nDataset information:")
print(df_eda.info())

print("\nFirst five records:")
print(df_eda.head())
```

```
# -----
```

3. NUMERICAL VARIABLES

```
# -----
```

```
numeric_columns = [
```

```
    "distance_km",
```

```
    "package_weight_kg",
```

```
    "delivery_rating",
```

```
    "delivery_cost"
```

```
]
```

```
print("\nDescriptive statistics:")
```

```
print(df_eda[numeric_columns].describe())
```

```
# -----
```

4. CENTRAL TENDENCY

```
# -----
```

```
central_tendency = pd.DataFrame({
```

```
    "Mean": df_eda[numeric_columns].mean(),
```

```
    "Median": df_eda[numeric_columns].median(),
```

```
    "Standard Deviation": df_eda[numeric_columns].std()
```

```
})
```

```
print("\nCentral tendency:")  
print(central_tendency)
```

```
# -----  
# 5. DISTRIBUTION OF DELIVERY DISTANCE  
# -----
```

```
plt.figure(figsize=(8, 5))
```

```
sns.histplot(  
    df_eda["distance_km"],  
    bins=30,  
    kde=True  
)
```

```
plt.title("Distribution of Delivery Distance")  
plt.xlabel("Distance (km)")  
plt.ylabel("Number of Deliveries")  
plt.tight_layout()  
plt.show()
```

```
# -----  
# 6. DISTRIBUTION OF PACKAGE WEIGHT  
# -----
```

```
plt.figure(figsize=(8, 5))
```

```
sns.histplot(  
    df_eda["package_weight_kg"],  
    bins=30,  
    kde=True  
)
```

```
plt.title("Distribution of Package Weight")  
plt.xlabel("Package Weight (kg)")  
plt.ylabel("Number of Deliveries")  
plt.tight_layout()  
plt.show()
```

```
# -----  
# 7. DISTRIBUTION OF DELIVERY RATING  
# -----
```

```
plt.figure(figsize=(8, 5))
```

```
sns.countplot(  
    data=df_eda,  
    x="delivery_rating"  
)
```

```
plt.title("Distribution of Delivery Ratings")
plt.xlabel("Delivery Rating")
plt.ylabel("Number of Deliveries")
plt.tight_layout()
plt.show()
```

```
# -----
# 8. DISTRIBUTION OF DELIVERY COST
# -----
```

```
plt.figure(figsize=(8, 5))
```

```
sns.histplot(
    df_eda["delivery_cost"],
    bins=30,
    kde=True
)
```

```
plt.title("Distribution of Delivery Cost")
plt.xlabel("Delivery Cost")
plt.ylabel("Number of Deliveries")
plt.tight_layout()
plt.show()
```

```
# -----
```

```
# 9. DELIVERY COST BOX PLOT
```

```
# -----
```

```
plt.figure(figsize=(8, 5))
```

```
sns.boxplot(  
    y=df_eda["delivery_cost"]  
)
```

```
plt.title("Box Plot of Delivery Cost")
```

```
plt.ylabel("Delivery Cost")
```

```
plt.tight_layout()
```

```
plt.show()
```

```
# -----
```

```
# 10. DELAY RATE BY DELIVERY MODE
```

```
# -----
```

```
delay_by_mode = (  
    df_eda.groupby("delivery_mode")["delayed"]  
    .apply(lambda x: (x == "yes").mean() * 100)  
    .sort_values(ascending=False)  
)
```

```
print("\nDelay rate by delivery mode:")
```

```
print(delay_by_mode)
```

```
plt.figure(figsize=(8, 5))
```

```
sns.barplot(  
    x=delay_by_mode.index,  
    y=delay_by_mode.values  
)
```

```
plt.title("Delay Rate by Delivery Mode")
```

```
plt.xlabel("Delivery Mode")
```

```
plt.ylabel("Delay Rate (%)")
```

```
plt.xticks(rotation=20)
```

```
plt.tight_layout()
```

```
plt.show()
```

```
# -----
```

```
# 11. DELAY RATE BY WEATHER
```

```
# -----
```

```
delay_by_weather = (
```

```
    df_eda.groupby("weather_condition")["delayed"]
```

```
    .apply(lambda x: (x == "yes").mean() * 100)
```

```
.sort_values(ascending=False)
)

print("\nDelay rate by weather condition:")
print(delay_by_weather)

plt.figure(figsize=(8, 5))

sns.barplot(
    x=delay_by_weather.index,
    y=delay_by_weather.values
)

plt.title("Delay Rate by Weather Condition")
plt.xlabel("Weather Condition")
plt.ylabel("Delay Rate (%)")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```

```
# -----
```

```
# 12. DELAY RATE BY DELIVERY PARTNER
```

```
# -----
```

```
delay_by_partner = (
```

```
df_eda.groupby("delivery_partner")["delayed"]  
.apply(lambda x: (x == "yes").mean() * 100)  
.sort_values(ascending=False)  
)
```

```
print("\nDelay rate by delivery partner:")  
print(delay_by_partner)
```

```
plt.figure(figsize=(10, 5))
```

```
sns.barplot(  
    x=delay_by_partner.values,  
    y=delay_by_partner.index  
)
```

```
plt.title("Delay Rate by Delivery Partner")  
plt.xlabel("Delay Rate (%)")  
plt.ylabel("Delivery Partner")  
plt.tight_layout()  
plt.show()
```

```
# -----  
# 13. DELAY RATE BY REGION  
# -----
```

```
delay_by_region = (  
    df_eda.groupby("region")["delayed"]  
    .apply(lambda x: (x == "yes").mean() * 100)  
    .sort_values(ascending=False)  
)
```

```
print("\nDelay rate by region:")  
print(delay_by_region)
```

```
plt.figure(figsize=(8, 5))
```

```
sns.barplot(  
    x=delay_by_region.index,  
    y=delay_by_region.values  
)
```

```
plt.title("Delay Rate by Region")  
plt.xlabel("Region")  
plt.ylabel("Delay Rate (%)")  
plt.tight_layout()  
plt.show()
```

```
# -----  
# 14. DELAY RATE BY VEHICLE TYPE  
# -----
```

```
delay_by_vehicle = (  
    df_eda.groupby("vehicle_type")["delayed"]  
    .apply(lambda x: (x == "yes").mean() * 100)  
    .sort_values(ascending=False)  
)
```

```
print("\nDelay rate by vehicle type:")  
print(delay_by_vehicle)
```

```
# -----  
# 15. CORRELATION MATRIX  
# -----
```

```
correlation_matrix = df_eda[numeric_columns].corr()
```

```
print("\nCorrelation matrix:")  
print(correlation_matrix.round(3))
```

```
# -----  
# 16. CORRELATION HEATMAP  
# -----
```

```
plt.figure(figsize=(8, 6))
```

```
sns.heatmap(  
    correlation_matrix,  
    annot=True,  
    cmap="coolwarm",  
    fmt=".2f",  
    center=0  
)
```

```
plt.title("Correlation Matrix of Logistics Variables")  
plt.tight_layout()  
plt.show()
```

```
# -----  
# 17. DISTANCE VS DELIVERY COST  
# -----
```

```
plt.figure(figsize=(9, 6))
```

```
sns.scatterplot(  
    data=df_eda,  
    x="distance_km",  
    y="delivery_cost",  
    alpha=0.5  
)
```

```
plt.title(  
    "Relationship Between Delivery Distance and Delivery Cost"  
)  
plt.xlabel("Delivery Distance (km)")  
plt.ylabel("Delivery Cost")  
plt.tight_layout()  
plt.show()
```

```
# -----  
# 18. PACKAGE WEIGHT VS DELIVERY COST  
# -----
```

```
plt.figure(figsize=(9, 6))
```

```
sns.scatterplot(  
    data=df_eda,  
    x="package_weight_kg",  
    y="delivery_cost",  
    alpha=0.5  
)
```

```
plt.title(  
    "Relationship Between Package Weight and Delivery Cost"  
)
```

```
plt.xlabel("Package Weight (kg)")
plt.ylabel("Delivery Cost")
plt.tight_layout()
plt.show()
```

```
# -----
```

```
# 19. OVERALL DELAY RATE
```

```
# -----
```

```
overall_delay_rate = (
    (df_eda["delayed"] == "yes").mean() * 100
)
```

```
print(
    "\nOverall delay rate:",
    round(overall_delay_rate, 2),
    "%"
)
```

```
# -----
```

```
# 20. DELIVERY STATUS DISTRIBUTION
```

```
# -----
```

```
status_distribution = (
```

```
df_eda["delivery_status"]  
.value_counts(normalize=True) * 100  
)
```

```
print("\nDelivery status distribution:")  
print(status_distribution.round(2))
```

```
# -----  
# 21. MODE PERFORMANCE SUMMARY  
# -----
```

```
mode_summary = df_eda.groupby("delivery_mode").agg(  
    Average_Cost=("delivery_cost", "mean"),  
    Average_Rating=("delivery_rating", "mean"),  
    Average_Distance=("distance_km", "mean")  
)
```

```
mode_summary["Delay_Rate"] = (  
    df_eda.groupby("delivery_mode")["delayed"]  
    .apply(lambda x: (x == "yes").mean() * 100)  
)
```

```
print("\nDelivery mode performance:")  
print(mode_summary.round(2))
```

```
# -----
```

```
# 22. FINAL OUTPUT
```

```
# -----
```

```
print("\n===== PROJECT 3 SUMMARY  
=====")
```

```
print("Number of records:", len(df_eda))
```

```
print("Number of analytical variables:", len(df_eda.columns))
```

```
print(  
    "Average distance:",  
    round(df_eda["distance_km"].mean(), 2)  
)
```

```
print(  
    "Average package weight:",  
    round(df_eda["package_weight_kg"].mean(), 2)  
)
```

```
print(  
    "Average rating:",  
    round(df_eda["delivery_rating"].mean(), 2)  
)
```

```
print(  
    "Average delivery cost:",  
    round(df_eda["delivery_cost"].mean(), 2)  
)
```

```
print(  
    "Overall delay rate:",  
    round(overall_delay_rate, 2),  
    "%"  
)
```

```
print("\nProject 3 EDA completed.")
```